JAVA SNACKS – FLAGYL
(Saturday, May 2, 2026)

Instructions:

1. Avoid using Artificial Intelligence(AI) tools.

2. Submission must occur before class (8:30pm – Monday, May 4, 2026).

3. Each question is to be in a different file.

Submission Guideline: It is safe to assume you have created a repo called JavaTasks(or its equivalent).

Add to the structures and save all of today's tasks inside it

Push to git.

1. Write a method: public static boolean isPrime(int n) that returns true if n is prime. Use this method in a program that finds all twin primes less than 1,200. Twin primes are pairs of primes that differ by 2 (e.g., 3 and 5, 5 and 7).
2. Write a program to play a game of craps. Roll two dice. If the sum is 7 or 11 (natural), you win. If the sum is 2, 3, or 12 (craps), you lose. Otherwise a point is established — keep rolling until you roll the point (win) or a 7 (lose). Display all rolls and the final result.
3. Write a program that simulates the craps game from the previous question 15,000 times and displays the number of winning games.
4. Write a program that uses System.currentTimeMillis() to compute and display the current date and time in the format: Current date and time is [Month Day, Year HH:MM:SS].
5. Write a method: public static double m(int i) that returns an approximation of Pi using: Pi ≈ 4(1 − 1/3 + 1/5 − 1/7 + ... + (−1)^(i+1) / (2i−1)). Write a test program that displays a table of m(i) for i = 1, 101, 201, 301, ..., 901.
6. Write a program that reads student scores, finds the best score, and assigns grades: A if score ≥ best−5, B if score ≥ best−10, C if score ≥ best−15, D if score ≥ best−20, F otherwise. Prompt the user to enter the number of students and all scores, then display each student's score and grade.
7. Write a program that reads 10 integers and displays them in the reverse order in which they were entered.
8. Write a program that reads integers between 1 and 50 until 0 is entered. Use an array of 50 integers to count the occurrences of each number. Display each number that appears at least once, along with its count, using 'time' for singular and 'times' for plural.
9. Write a program that reads an unspecified number of scores (ending with a negative number, max 100 scores) and determines how many scores are above or equal to the average and how many are below it.
10. Write a program that reads ten integers and displays the count of even numbers and the count of odd numbers. Assume input ends with 0.

Hint: Write a test program means write a main method
Write a test method means write a test file

Have fun guys!